

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 221 — Past Questions & Materials
by Renance

Course: GET 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — GET 221 ENGINEERING MATHEMATICS II

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

SECOND SEMESTER: 2023/2024 SESSION EXAMINATION

COURSE TITLE: ENGINEERING MATHEMATICS I Credit units: 3

COURSE CODE: GET 221 Time Allowed: 3 Hours.

INSTRUCTIONS: Answer five (5) Questions.

Question one is compulsory and then answer

at least one question from each section.

1. (u) Many practical problems in Engineering give rise to second order differential equations of the form:

$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = R(x)$

Which is a complementary solution with real and different roots.

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roots to auxiliary equation.

(a) With the aid of equations and sketched graphs. describe the following:

(a) Over damped.

(b) Critical damped. answer

rv ; t= bte : : 1

(c) Under damped. ;

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(a) Solve the equation, $\frac{d^2y}{dx^2} + 4y = 10 \sin x$;

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L.(b) If $F(r) = x^2y + 2yz + 3z^2x$ evaluate

"f= d ae , by

the distance between A(0,0,0) and B(12,3). use the formula

distance =

(b) Along the straight line C; from (0,0,0) to (1,2,3) find the distance =

? distance =

(b). Along the straight line C2 from (1,0,0) to (1,2,0) =

(b), Along the straight line C3 from (1,2,0) to (1,2,3)

(b). Along the straight line C4 joining (0,0,0) to (1,2,3) find

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SECTION A

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2. (a) A mass weighing 2 pounds, stretches a spring 6 inches. At, $t = 0$ the mass is released

; $v_0 = 3$ ft/s. Determine the

position of the mass at any time t .

equation of free motion and find the roots of the equation.

(b) Interpret and solve the initial-value problem.

$10x'' + 5x = 0$

$x(0) = 1/2, x'(0) = 0$

$x'' + 1.25x = 0$

$x(0) = 1/2, x'(0) = 0$

(c) Evaluate, $L = \int_0^1 (x^2 + x) dx$

2

d^2y/dx^2

3. (a) Solve the equation, $y'' + 4y = 0$

ty

$Be^{2y} = a^2 \sin y$

(b) Solve the equation, $y'' + 2y = x'$, given that when $x = 0$, $y = 0$ and $y' = 1$

(c) Solve the equation, $y'' + 4y = 13e^{2x}$, given that when $x = 0$, $y = 0$ and $y' = 1$

dx^2

4. (a) Find the volume of the solid bounded by the planes $z = 0$, $x = 1$, $x = 4$, $y = 2$, $y = 5$ and the surface $z = x + y$. Also draw the solid in three dimensional direction,

by the planes

$5 \leq x \leq 10$ and $0 \leq y \leq 10$

$y = 0$, $z = 0$ and $z = 10$

A solid is bounded by the planes $x = 0$, $y = 0$, $z = 0$

and the surface $z = 10 - x - y$

(c) A mass suspended from a spring performs vertical oscillations and the displacement x (cm) of the mass at time t (s) is given by

1 d^*x

$\text{rer} | = -48x.$

If x - and $= 0$ when $t = 0$, determine the period and amplitude of the oscillations.

SECTION B

5.(a) If $A = (y^* - x^2z?)i + (x+y) - (x^*yz)k$ Determine curl A at the point (1,3

$=3$ To

(b) If $F = xy7i - 2yzj + xyzk$ compute $\text{div } F$ at (1-173)

(c) F-valueate $1 = (x+y) dx$ from A(0.1) to BiO,-1) along semi-circle $x^{\circ} + y? = 1$, for $x > 0$

(d) | valuate $lxy44$, where C is the right half circle $x^* + y" = 16$ traced out in acounter Clockwisg «>
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direcuon. Given that $x=4\text{cost}$, $y=4\text{sint}$ ie "Fell ni [6 at nteo

j a ;

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6.(a) Anxty + (xy +yz)j +xz7k (1.2.1) find curl curl A.

(b) Determine if the following vector field is conservative

$F(x,y) = (2xe\% + tye); + (x3e FFy)j$

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(c) Determine if the integral $\int f dx$, converges or diverges.

QO ($x=2$).?

(d) Evaluate the line integra! $I = \int xydx + (2x-y)dy$ round the region bounded by the curve $y=x'$ and $x -y^*$ by the use of Green's theorem

7.(a) Evaluate $\int ydx$ where C is the circle $x^{\circ} + y^2 = 4$

(b) If $F(r) = x^2y'i + y^3zj + z'k$ evaluate $\int F \cdot dr$ along the curve $x=2U'$, $y=3U$ $z=U?$ between A(2.-3.-1) and B(2.3,1)

(c) If $F = x?yzi + xyz?j + y*zk$ determine curl curl F at (2,1,1)

(d) Determine curl F at the point (2,0,3) Given that

$F = Ze7^{*"}1 + 2xzCosyj + (x+2y)k$

Paper 2 of 3 — GET 221

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, FFURUN



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COLLEGE OF ENGINEERING TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING
SECOND SEMESTER; 2019/2020 SESSION EXAMINATION

COURSE TITLE: ENGINEERING MATHEMATICS II

COURSE CODE: GET221

Credit Units: 3

Time Allowed: 3 hours

INSTRUCTION: Answer AT LEAST TWO (2) Questions from each session.

COURSE TITLE: ENGINEERING MATHEMATICS II

Section 1

1. Determine whether the following integrals converge or diverge

(a) $\int_1^{\infty} \frac{4}{x^3} dx$

(b) $\int_{-\infty}^1 e^x dx$

(c) $\int_0^4 \frac{1}{(x-2)^2} dx$

2. Evaluate $\int_C \sin(\pi y) dy + yx^2 dx$ where C is the line segment from (0, 2) to (1, 4)

3. Evaluate $\int_C \frac{1}{\sqrt{x^2+y^2}} dS$, where C is the right half of the circle, $x^2 + y^2 = 16$ traced out in a counter clockwise direction

4. Evaluate $\oint_C y^3 dx - x^3 dy$, where C is the positively oriented circle of radius 2 centered at the origin

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Section 2

5. ✓ Solve by elimination approach the initial value problem of the system of differential equation given below

$$x_1' + 2x_1 - x_2 = 1 + e^{-t}$$

$$x_2' + x_1 + 2x_2 = 3 \quad \text{with } x_1(0) = 5/2 \text{ and } x_2(0) = -1/2$$

6. (a) Find the complementary function (CI) and the particular integral (PI) for the differential equation given below;

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2$$

(b) Evaluate $\int_1^2 \int_0^3 \int_0^1 (p^2 + q^2 - r^2) dp dq dr$

(c) Find the area under the curve $y = 4\sin \frac{x}{2}$ between $x = \frac{\pi}{3}$ and $x = \pi$

7. ✓ Solve for the roots of the following equations

(a) $x^5 - 3x^4 + 3x^3 - 9x^2 - 4x + 12 = 0$

(b) $x^3 - 6x^2 + 13x - 10 = 0$

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Paper 3 of 3 — GET 221 PAST QUESTIONS

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, FFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER; 2020/2021 ACADEMIC SESSION EXAMINATION

COURSE TITLE: ENGINEERING MATHEMATICS II

COURSE CODE: GET221

Credit Units: 3

Time Allowed: 3 hours

INSTRUCTION: Answer Five (5) questions; AT LEAST TWO (2) Questions from each section.

Section A

1. Solve for the roots of the following equations

(a) $x^3 - 6x^2 + 13x - 10 = 0$

(b) $x^5 - 3x^4 + 3x^3 - 9x^2 - 4x + 12 = 0$

2. (a) Evaluate $\int_1^2 \int_0^3 \int_0^1 (p^2 + q^2 - r^2) dp dq dr$

(b) Find the area under the curve $y = 4\sin \frac{x}{2}$ between $x = \frac{\pi}{3}$ and $x = \pi$

(c) Find the complementary function (CF) and the particular integral (PI) for the differential equation given below;

$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2$

3. Solve by elimination approach the initial value problem of the system of differential equation given below

$x_2' + x_1 + 2x_2 = 3$

$x_1' + 2x_1 - x_2 = 1 + e^{-t}$ with $x_1(0) = 5/2$ and $x_2(0) = -1/2$

$\int_1^\infty \frac{1}{x} dx$

$\int_1^\infty \frac{1}{x^2} dx$

Section B

4. a) Evaluate

$\int_C xy^2 ds$ where C is the right half of a circle, $x^2 + y^2 = 16$ traced out in a counter clockwise direction.

b) Find the convergence or divergence of the following integrals.

i. $\int_1^\infty \frac{1}{x} dx$

ii. $\int_1^\infty \frac{1}{x^3} dx$

iii. $\int_{-\infty}^0 e^x dx$

$\frac{2^x}{\ln 2} + \frac{1}{2} 2^x$